

习 题 9.5

1. 求下列微分方程的通解:

$$(1) y'' - 3y' + 2y = 0; \quad (2) 4y'' + 5y' + y = 0;$$

$$(3) y'' + 6y' + 9y = 0; \quad (4) y'' + 4y' + 5y = 0;$$

$$(5) y'' - y' + 2y = 0; \quad (6) y''' + 2y'' - y' = 0.$$

2. 求下列初值问题的解:

$$(1) \begin{cases} y'' + 2y' + 4y = 0, \\ y(0) = 1, y'(0) = -1; \end{cases} \quad (2) \begin{cases} 4y'' + 4y' + y = 0, \\ y(0) = 0, y'(0) = 2. \end{cases}$$

3. 用待定系数法求下列方程的特解:

$$(1) y'' - 3y' + 5y = 6; \quad (2) y'' + 3y' = 6e^{2x};$$

$$(3) y'' - 9y' + 20y = x + 1; \quad (4) y'' + y = 4\sin x;$$

$$(5) y'' - 3y' + 2y = x \cos x; \quad (6) y'' - 9y = e^{3x} \cos x;$$

$$(7) y'' - y = 2e^x - x^2; \quad (8) y'' + y' = \sin 4x - 2\sin 2x.$$

$$(5). \lambda^2 - \lambda + 2 = 0, \quad \lambda = \frac{1 \pm \sqrt{-7}}{2} = \frac{1 \pm \sqrt{7}i}{2}$$

$$C_1 e^{\frac{1}{2}x} (\cos \frac{\sqrt{7}}{2}x + \sin \frac{\sqrt{7}}{2}x)$$

$$(6) \lambda^3 + 2\lambda^2 - \lambda = 0, \quad \lambda(\lambda^2 + 2\lambda - 1) = 0, \quad \lambda_1 = 0, \lambda_{2,3} = \frac{-2 \pm \sqrt{8}}{2} = -1 \pm \sqrt{2}$$

$$C_1 + C_2 e^{-1+\sqrt{2}} + C_3 e^{-1-\sqrt{2}}$$

$$2. (1) \lambda^2 + 2\lambda + 4 = 0, \quad \Delta = 4 - 16 = -12$$

$$\frac{-2 \pm \sqrt{-12}}{2} = -1 \pm \sqrt{3}i$$

$$\begin{cases} y = e^{-x} (C_1 \cos \sqrt{3}x + C_2 \sin \sqrt{3}x) \\ y' = e^{-x} (\sqrt{3}C_2 \cos \sqrt{3}x - \sqrt{3}C_1 \sin \sqrt{3}x - C_1 \cos \sqrt{3}x - C_2 \sin \sqrt{3}x) \end{cases}$$

$$y(0) = C_1 = 1, \quad y'(0) = \sqrt{3}C_2 - C_1 = -1, \quad C_2 = 0$$

$$\therefore y = e^{-x} (\cos \sqrt{3}x)$$

$$1. (1) y^2 - 3\lambda + 2 = 0, \quad \lambda_1 = 1, \lambda_2 = 2$$

$$C_1 e^x + C_2 e^{2x}$$

$$(2) 4\lambda^2 + 5\lambda + 1 = 0$$

$$\Delta = 25 - 16 = 9$$

$$\lambda_1 = -\frac{1}{4}, \lambda_2 = -1$$

$$C_1 e^{-\frac{1}{4}x} + C_2 e^{-x}$$

$$(3) \lambda^2 + 6\lambda + 9 = 0, \quad \lambda_1 = \lambda_2 = -3$$

$$C_1 e^{-3x} + C_2 x e^{-3x}$$

$$(4) \lambda^2 + 4\lambda + 5 = 0, \quad \Delta = 16 - 20 = -4$$

$$\frac{-4 \pm \sqrt{-4}}{2} = -2 \pm i$$

$$C_1 e^{-2x} \cos x + C_2 e^{-2x} \sin x$$

$$(2) 4\lambda^2 + 4\lambda + 1 = 0, \quad (2\lambda + 1)^2 = 0, \quad \lambda_{1,2} = -\frac{1}{2}$$

$$y = (C_1 + C_2 x) e^{-\frac{1}{2}x}$$

$$y' = C_2 e^{-\frac{1}{2}x} + (C_1 + C_2 x) \cdot (-\frac{1}{2}) \cdot e^{-\frac{1}{2}x}$$

$$y(0) = C_1 = 0, \quad y'(0) = C_2 = 2.$$

$$y = 2x e^{-\frac{1}{2}x}$$

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(5) $\lambda^2 - 3\lambda + 2 = 0, \lambda_1 = 1, \lambda_2 = 2.$

$$y = (a_1x + a_0) \cos x + (b_1x + b_0) \sin x$$

$$y' = a_1 \cos x - (a_1x + a_0) \sin x + b_1 \sin x + (b_1x + b_0) \cos x = (a_1 + b_0 + b_1x) \cos x + (b_1 - a_1x - a_0) \sin x$$

$$y'' = -a_1 \sin x - a_1 \sin x - (a_1x + a_0) \cos x + b_1 \cos x + b_1 \cos x - (b_1x + b_0) \sin x = (-2a_1 - b_1x - b_0) \sin x + (2b_1 - a_1x - a_0) \cos x$$

$$(-2a_1 - b_1x - b_0 - 3b_1 + 3a_1x + 3a_0 + 2b_1x + 2b_0) \sin x + (2b_1 - a_1x - a_0 - 3a_1 - 3b_0 - 3b_1x + 2a_1x + 2a_0) \cos x$$

$$= [(b_1 + 3a_1)x + 3a_0 - 2a_1 + b_0 - 3b_1] \sin x + [(a_1 - 3b_1)x + a_0 - 3a_1 + 2b_1 - 3b_0] \cos x$$

$$\begin{cases} b_1 + 3a_1 = 0 \\ 3a_0 - 2a_1 + b_0 - 3b_1 = 0 \\ a_1 - 3b_1 = 1 \\ a_0 - 3a_1 + 2b_1 - 3b_0 = 0 \end{cases} \Rightarrow \begin{cases} a_1 = \frac{1}{10} \\ b_1 = -\frac{3}{10} \\ a_0 = -\frac{3}{25} \\ b_0 = \frac{-13}{50} \end{cases}$$

$$\therefore y = (\frac{1}{10}x - \frac{3}{25}) \cos x + (-\frac{3}{10}x - \frac{13}{50}) \sin x$$

(6) $\lambda^2 - 9 = 0, \lambda = \pm 3$

$$y = e^{3x} \cdot (A \cos x + B \sin x)$$

$$y' = 3e^{3x} \cdot (A \cos x + B \sin x) + e^{3x} \cdot (B \cos x - A \sin x)$$

$$= e^{3x} [(3A + B) \cos x + (3B - A) \sin x]$$

3. (1) $\lambda^2 - 3\lambda + 5 = 0$, 0 不是特征根

$$Q(x) = C_1, \text{ 代入原方程}$$

$$5C_1 = 6, C_1 = \frac{6}{5}$$

$$\text{特解: } y = \frac{6}{5}$$

(3) $\lambda^2 - 9\lambda + 20 = 0, \lambda_1 = 4, \lambda_2 = 5,$

$$Q_n(x) = a_1x + a_0$$

$$-9a_1 + 20a_1x + 20a_0 = x + 1$$

$$a_1 = \frac{1}{20}$$

$$a_0 = \frac{29}{400}$$

(2) $\lambda^2 + 3\lambda = 0, \lambda_1 = 0, \lambda_2 = -3$, 2 不是特征根

$$Q(x) = ae^{ax}, \text{ 代入原方程}$$

$$(a^2 + 3a) \cdot e^{ax} = 6e^{2x}$$

$$a = 2, 10a \cdot e^{2x} = 6e^{2x}, a = \frac{3}{5}$$

$$\text{特解 } y = \frac{3}{5}e^{2x}$$

(4) $\lambda^2 + 1 = 0, \lambda = \pm i,$

$$\text{设 } y = x \cdot (A \cos x + B \sin x)$$

$$y' = A \cos x + B \sin x + x \cdot (-B \cos x - A \sin x)$$

$$y'' = B \cos x - A \sin x + B \cos x - A \sin x + x \cdot (-B \sin x + A \cos x)$$

$$= 2(B \cos x - A \sin x) - x \cdot (B \sin x + A \cos x)$$

$$2(B \cos x - A \sin x) = 4 \sin x$$

$$\begin{cases} A = -2 \\ B = 0 \end{cases} \therefore y = -2x \cos x$$

$$a_0 = -\frac{6}{50} = -\frac{3}{25}$$

$$3b_0 = \frac{3}{25} - \frac{9}{10} = \frac{6}{50} - \frac{45}{50} = \frac{-39}{50}, b_0 = \frac{-13}{50}$$

$$y'' = 3e^{3x} [\quad] + e^{3x} [(3B-A)\cos x - (3A+B)\sin x]$$

$$= e^{3x} [(8A+6B)\cos x + (8B-6A)\sin x]$$

$$e^{3x} \cdot [(6B-A)\cos x + (-6A-B)\sin x] = e^{3x} \cos x$$

$$\begin{cases} 6B-A=1 \\ -6A-B=0 \end{cases} \quad \begin{cases} A=-\frac{1}{37} \\ B=\frac{6}{37} \end{cases}$$

$$\therefore y = \frac{1}{37} e^{3x} (-\cos x + 6\sin x)$$

(7) $y'' - y = 2e^x - x^2$;

(8) $y'' + y' = \sin 4x - 2\sin 2x$.

4. 对下列各方程, 写出其待定系数的特解形式:

(7). $\lambda^2 - 1 = 0, \lambda = \pm 1$

① $y'' - y = 2e^x$

$$y = ax \cdot e^x, y'' = a(x+2)e^x$$

$$\therefore 2ae^x = 2e^x, a=1$$

$$y = xe^x$$

② $y'' - y = -x^2, a_n(x) = b_2x^2 + b_1x + b_0$

$$\therefore a_n''(x) = 2b_2,$$

$$\therefore \begin{cases} -b_2 = -1 \\ -b_1 = 0 \\ 2b_2 - b_0 = 0 \end{cases} \quad \begin{cases} b_2 = 1 \\ b_1 = 0 \\ b_0 = 2 \end{cases}$$

$$\therefore a_n(x) = x^2 + 2$$

$$\therefore \text{通解 } y = xe^x + x^2 + 2$$

(8) $\lambda^2 + \lambda = 0, \lambda_1 = 0, \lambda_2 = -1$

① $y'' + y' = \sin 4x$

$$f(x) = A \cos 4x + B \sin 4x$$

$$f'(x) = -4A \sin 4x + 4B \cos 4x$$

$$f''(x) = -16A \cos 4x - 16B \sin 4x$$

$$\therefore (-16B - 4A) \sin 4x + (4B - 16A) \cos 4x = \sin 4x$$

$$\begin{cases} -16B - 4A = 1 \\ 4B - 16A = 0 \end{cases} \quad \begin{cases} A = -\frac{1}{68} \\ B = -\frac{1}{17} \end{cases}$$

$$\therefore f(x) = -\frac{1}{68} \cos 4x - \frac{1}{17} \sin 4x$$

② $y'' + y' = 2 \sin 2x$

$$g(x) = A \cos 2x + B \sin 2x$$

$$g'(x) = -2A \sin 2x + 2B \cos 2x$$

$$g''(x) = -4A \cos 2x - 4B \sin 2x$$

$$(2B - 4A) \cos 2x + (-2A - 4B) \sin 2x = 2 \sin 2x$$

$$\begin{cases} 2B - 4A = 0 \\ -2A - 4B = 2 \end{cases} \quad \begin{cases} A = -\frac{1}{5} \\ B = -\frac{2}{5} \end{cases}$$

$$g(x) = -\frac{1}{5} \cos 2x - \frac{2}{5} \sin 2x$$

